

Highlights

An Operator Splitting ADI Scheme for Two-Dimensional Tumor Angiogenesis

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- Two-dimensional tumor angiogenesis PDE system extended from the 1D reference model using Kronecker product spatial operators on a uniform Cartesian grid.
- Strang operator splitting decouples the stiff linear diffusion operator from the non-stiff nonlinear reaction-advection terms.
- Peaceman–Rachford ADI reduces the 2D implicit diffusion step to batched 1D tridiagonal solves.

An Operator Splitting ADI Scheme for Two-Dimensional Tumor Angiogenesis^{*}

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ABSTRACT

We present the mathematical formulation and numerical discretization of a two-dimensional extension of the tumor angiogenesis PDE system originally proposed in one spatial dimension. The model describes the spatio-temporal evolution of endothelial cell density, protease concentration, inhibitor concentration, and extracellular matrix density, coupled through diffusion, chemotaxis, haptotaxis, and nonlinear reaction kinetics. The numerical framework employs the Method of Lines for spatial semi-discretization with second-order central differences on a uniform Cartesian grid, where two-dimensional Laplacian and gradient operators are constructed via Kronecker products of one-dimensional operators with Neumann boundary conditions. Time integration is performed through Strang splitting, decoupling diffusion from reaction-advection substeps. The diffusion substep is solved by the Alternating Direction Implicit method of Peaceman–Rachford, reducing the two-dimensional implicit problem to independent batches of one-dimensional tridiagonal systems. The reaction-advection substep is treated explicitly with upwind differencing for the chemotactic flux. The complete algebraic formulation is presented in matrix-vector notation.

1. Introduction

Tumor angiogenesis, the formation of new blood vessels from pre-existing vasculature, is a fundamental process in cancer progression. Mathematical modeling of angiogenesis through systems of partial differential equations (PDEs) provides a quantitative framework for understanding the complex interactions among endothelial cells, proteolytic enzymes, angiogenic inhibitors, and the extracellular matrix. The one-dimensional PDE model analyzed in De Luca and Marcellino (2026); De Luca and Marcellino (2025) captures the essential dynamics of these interactions, and its numerical treatment has been studied with respect to convergence properties and discrete conservation laws.

The present work extends that framework to two spatial dimensions, which is necessary to capture biologically relevant phenomena such as branching morphogenesis, lateral migration, and the spatial heterogeneity of the tumor angiogenic factor (TAF) field. The transition from one to two dimensions introduces significant computational challenges: the number of degrees of freedom scales quadratically with the mesh resolution, the CFL stability constraints become more restrictive, and the linear algebra cost per time step increases substantially.

To address these challenges, we adopt a numerical strategy based on operator splitting combined with the Alternating Direction Implicit (ADI) method. This approach offers two fundamental advantages. First, Strang splitting decouples the stiff diffusion operator from the non-stiff reaction and advection terms, allowing each substep to be treated with an appropriate integrator. Second, the ADI method of Peaceman and Rachford Peaceman and Rachford (1955) reduces the two-dimensional implicit diffusion problem to independent batches of one-dimensional tridiagonal systems, which can be solved in $\mathcal{O}(M)$ operations per system via the Thomas algorithm. **This structure maps directly to the batched tridiagonal solver `gtsv2StridedBatch` available in the `cuSPARSE` library NVIDIA Corporation (2024), enabling efficient GPU parallelization without requiring global sparse factorizations.**

The rest of this paper is organized as follows. Section 2 presents the mathematical formulation of the two-dimensional angiogenesis model, including the governing equations, boundary and initial conditions, the TAF field, and the reformulation of the chemotactic flux terms. Section 3 develops the complete numerical discretization, proceeding

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from the spatial semi-discretization through the Strang splitting architecture, the ADI diffusion solver, and the explicit reaction-advection substep, concluding with the compact algebraic formulation and a summary of the computational properties.

2. Mathematical Background

We recall the one-dimensional PDE system studied in De Luca and Marcellino (2026), which constitutes the starting point for the two-dimensional extension. Let $\Omega_{1D} = [0, L_f]$ denote the spatial domain and $[t_0, T_f]$ the time interval. The model describes the evolution of four coupled fields: the endothelial cell (EC) density $C(x, t)$, the protease concentration $P(x, t)$, the angiogenic inhibitor concentration $I(x, t)$, and the extracellular matrix (ECM) density $F(x, t)$. The governing system reads

$$\begin{aligned}\frac{\partial C}{\partial t} &= d_C \frac{\partial^2 C}{\partial x^2} + \frac{\partial}{\partial x} \left(f_I \frac{\partial I}{\partial x} \right) - \frac{\partial}{\partial x} \left(f_F \frac{\partial F}{\partial x} \right) - \frac{\partial}{\partial x} \left(f_T \frac{\partial T}{\partial x} \right) + k_1 C(1 - C), \\ \frac{\partial P}{\partial t} &= d_P \frac{\partial^2 P}{\partial x^2} - k_3 P I + k_4 T C + k_5 T - k_6 P, \\ \frac{\partial I}{\partial t} &= d_I \frac{\partial^2 I}{\partial x^2} - k_3 P I, \\ \frac{\partial F}{\partial t} &= -k_2 P F,\end{aligned}\tag{1}$$

where the chemotactic and haptotactic sensitivity functions are

$$f_F = \alpha_1 C, \quad f_I = \alpha_2 C, \quad f_T = \frac{\alpha_3 C}{1 + \alpha_4 T},\tag{2}$$

and the TAF concentration $T(x)$ is prescribed as the steady-state profile $T(x) = \exp(-\epsilon^{-1}(L_f - x)^2)$. The system is subject to homogeneous Neumann boundary conditions

$$\left. \frac{\partial u}{\partial x} \right|_{x=0} = \left. \frac{\partial u}{\partial x} \right|_{x=L_f} = 0 \quad \text{for } u \in \{C, P, I, F\}.\tag{3}$$

2.1. Extension to two spatial dimensions

We extend system (1) to the rectangular domain $\Omega = [0, L_x] \times [0, L_y] \subset \mathbb{R}^2$ with boundary $\partial\Omega$. The four fields become functions of $\mathbf{x} = (x, y) \in \Omega$ and $t \in [t_0, T_f]$. The two-dimensional governing system takes the form

$$\begin{aligned}\frac{\partial C}{\partial t} &= d_C \Delta C + \nabla \cdot (f_I \nabla I) - \nabla \cdot (f_F \nabla F) - \nabla \cdot (f_T \nabla T) + k_1 C(1 - C), \\ \frac{\partial P}{\partial t} &= d_P \Delta P - k_3 P I + k_4 T C + k_5 T - k_6 P, \\ \frac{\partial I}{\partial t} &= d_I \Delta I - k_3 P I, \\ \frac{\partial F}{\partial t} &= -k_2 P F,\end{aligned}\tag{4}$$

$(\mathbf{x}, t) \in \Omega \times [t_0, T_f],$

where

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}, \quad \nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right)^\top\tag{5}$$

are the Laplacian and the gradient, respectively. The sensitivity functions retain their algebraic form (2), now evaluated on the two-dimensional fields.

The TAF concentration is extended to a two-dimensional steady-state profile modeling a localized tumor source at $(L_x, L_y/2)$:

$$T(\mathbf{x}) = \exp\left(-\epsilon^{-1}[(x - L_x)^2 + (y - L_y/2)^2]\right), \quad (6)$$

with gradient

$$\nabla T = -\frac{2}{\epsilon} \begin{pmatrix} x - L_x \\ y - L_y/2 \end{pmatrix} T(\mathbf{x}). \quad (7)$$

The no-flux boundary conditions generalize to $\nabla u \cdot \mathbf{n}|_{\partial\Omega} = 0$ for all $u \in \{C, P, I, F\}$, where $\mathbf{n}$ is the outward unit normal. We prescribe a C^∞ sigmoid profile for the EC density, localized near $x = 0$:

$$C(\mathbf{x}, 0) = \frac{C_0}{2} \left(1 - \tanh\left(\frac{x - a}{\sigma}\right)\right), \quad (8)$$

where $C_0 > 0$ is the maximum density, $a > 0$ controls the front position, and $\sigma > 0$ the transition width. The remaining fields are initialized with smooth deterministic perturbations:

$$\begin{aligned} P(\mathbf{x}, 0) &= 0.1 + 0.01 \cos(2\pi x) \cos(2\pi y), \\ I(\mathbf{x}, 0) &= 0.1 + 0.005 \cos(4\pi x) \cos(4\pi y), \\ F(\mathbf{x}, 0) &= 1.0 + 0.01 \cos(\pi x) \cos(\pi y). \end{aligned} \quad (9)$$

The cosine perturbations satisfy the Neumann conditions and provide a mesh-independent definition. Following the approach in De Luca and Marcellino (2026), we reformulate the taxis terms in the EC equation by expanding the flux divergences using the product rule and definitions (2):

$$\nabla \cdot (f_I \nabla I) = \alpha_2 (\nabla C \cdot \nabla I + C \Delta I), \quad \nabla \cdot (f_F \nabla F) = \alpha_1 (\nabla C \cdot \nabla F + C \Delta F). \quad (10)$$

Considering the TAF chemotaxis, introducing the auxiliary potential

$$\varphi(\mathbf{x}) = \alpha_4^{-1} \log(1 + \alpha_4 T(\mathbf{x})),$$

so that

$$\nabla \varphi = \nabla T / (1 + \alpha_4 T),$$

we obtain

$$\nabla \cdot (f_T \nabla T) = \alpha_3 (\nabla C \cdot \nabla \varphi + C \Delta \varphi).$$

Substituting into the first equation of (4), the EC equation takes the advection-reaction-diffusion form

$$\frac{\partial C}{\partial t} = d_C \Delta C + \mathbf{v} \cdot \nabla C + (\nabla \cdot \mathbf{v}) C + k_1 C(1 - C), \quad (11)$$

where the advection velocity field and its divergence are

$$\mathbf{v} = \alpha_2 \nabla I - \alpha_1 \nabla F - \alpha_3 \nabla \varphi, \quad \nabla \cdot \mathbf{v} = \alpha_2 \Delta I - \alpha_1 \Delta F - \alpha_3 \Delta \varphi. \quad (12)$$

This reformulation splits the role of C as a transported quantity from the roles of I , F , and T as generators of the velocity field, and suggests us a suitable upwind discretizaion. Defining the solution vector $\mathbf{u} = (C, P, I, F)^\top$, system (4) is written as

$$\frac{\partial \mathbf{u}}{\partial t} = \underbrace{\mathcal{A}\mathbf{u}}_{\text{diffusion}} + \underbrace{\mathcal{F}(\mathbf{u})}_{\text{reaction + advection}}, \quad (13)$$

where $\mathcal{A}\mathbf{u} = (d_C \Delta C, d_P \Delta P, d_I \Delta I, 0)^\top$ is the linear diffusion operator and $\mathcal{F}(\mathbf{u})$ collects all nonlinear terms.

2.2. Conservation properties of the continuous system

Integrating each equation of (4) over Ω and using the Neumann conditions with the divergence theorem to eliminate all flux terms, we obtain the mass evolution laws

$$\begin{aligned} \frac{d}{dt} \int_{\Omega} C \, d\mathbf{x} &= \int_{\Omega} k_1 C(1 - C) \, d\mathbf{x}, & \frac{d}{dt} \int_{\Omega} P \, d\mathbf{x} &= \int_{\Omega} [-k_3 PI + k_4 TC + k_5 T - k_6 P] \, d\mathbf{x}, \\ \frac{d}{dt} \int_{\Omega} I \, d\mathbf{x} &= -k_3 \int_{\Omega} PI \, d\mathbf{x}, & \frac{d}{dt} \int_{\Omega} F \, d\mathbf{x} &= -k_2 \int_{\Omega} PF \, d\mathbf{x}. \end{aligned} \quad (14)$$

A key combined quantity is the protease-inhibitor balance. Defining $Q(t) = \int_{\Omega} [P - I] \, d\mathbf{x}$ and using the evolution equations for P and I in (4), we obtain

$$\frac{d}{dt} Q(t) = \int_{\Omega} \left[\frac{\partial P}{\partial t} - \frac{\partial I}{\partial t} \right] d\mathbf{x} = \int_{\Omega} [-k_3 PI + k_4 TC + k_5 T - k_6 P - (-k_3 PI)] d\mathbf{x} = \int_{\Omega} [k_4 TC + k_5 T - k_6 P] d\mathbf{x}, \quad (15)$$

where the diffusion terms vanish by the Neumann conditions and the $k_3 PI$ contributions cancel exactly: $-k_3 \int_{\Omega} PI \, d\mathbf{x} - (-k_3 \int_{\Omega} PI \, d\mathbf{x}) = 0$. This identity provides a diagnostic for the numerical scheme.

that is essential for convergence studies across different grid resolutions.

3. Numerical Discretization

In this section we provide a detailed discussion of adopted numerical schema. First, we introduce the semi-discretization approach, second, we address the full numerical scheme by introduced the splitting operator-based method.

3.1. Spatial discretization

We introduce a uniform Cartesian grid on $\Omega = [0, L_x] \times [0, L_y]$,

$$\Omega_h = \{(x_i, y_j) : x_i = (i-1)h_x, y_j = (j-1)h_y, i = 1, \dots, M_x, j = 1, \dots, M_y\}, \quad (16)$$

with mesh sizes $h_x = L_x/(M_x-1)$ and $h_y = L_y/(M_y-1)$, yielding $M = M_x M_y$ degrees of freedom per scalar variable and a total system dimension of $4M$. For any sufficiently smooth function $u \in C^4(\bar{\Omega})$ restricted to the grid as $u_{i,j} = u(x_i, y_j)$, the standard Taylor expansion about the point (x_i, y_j) gives

$$u(x_i \pm h_x, y_j) = u_{i,j} \pm h_x u_x|_{i,j} + \frac{h_x^2}{2} u_{xx}|_{i,j} \pm \frac{h_x^3}{6} u_{xxx}|_{i,j} + \frac{h_x^4}{24} u_{xxxx}|_{\xi_i^{\pm}, y_j} \quad (17)$$

for some $\xi_i^{\pm}$ between x_i and $x_i \pm h_x$. Summing the forward and backward expansions and solving for u_{xx} yields the central difference approximation to the second derivative:

$$D_{xx} u_{i,j} := \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h_x^2} = u_{xx}|_{i,j} + \frac{h_x^2}{12} u_{xxxx}|_{\bar{\xi}_i, y_j}, \quad (18)$$

where $\bar{\xi}_i \in (x_{i-1}, x_{i+1})$. The remainder is uniformly bounded by $(h_x^2/12) \|u_{xxxx}\|_{L^\infty(\Omega)}$, confirming $\mathcal{O}(h_x^2)$ accuracy. Analogously, the central difference for the second derivative in y satisfies

$$D_{yy} u_{i,j} := \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{h_y^2} = u_{yy}|_{i,j} + \frac{h_y^2}{12} u_{yyyy}|_{x_i, \bar{\eta}_j}, \quad (19)$$

with $\bar{\eta}_j \in (y_{j-1}, y_{j+1})$.

Subtracting the two Taylor expansions in (17) and dividing by $2h_x$ provides the central difference for the first derivative:

$$D_x u_{i,j} := \frac{u_{i+1,j} - u_{i-1,j}}{2h_x} = u_x|_{i,j} + \frac{h_x^2}{6} u_{xxx}|_{\bar{\xi}_i, y_j}, \quad (20)$$

with $\tilde{\xi}_i \in (x_{i-1}, x_{i+1})$, and similarly $D_y u_{i,j} = u_y|_{i,j} + (h_y^2/6) u_{yyy}|_{x_i, \tilde{\eta}_j}$.

At the domain boundaries, the homogeneous Neumann condition $\nabla u \cdot \mathbf{n}|_{\partial\Omega} = 0$ is imposed through reflecting ghost nodes. Consider the left boundary $i = 1$: the condition $u_x(0, y_j) = 0$ implies, via a Taylor expansion of $u_{0,j}$ about $x_1 = 0$,

$$u_{0,j} = u(x_1 - h_x, y_j) = u_{1,j} - h_x \underbrace{u_x|_{1,j}}_{=0} + \frac{h_x^2}{2} u_{xx}|_{1,j} + \mathcal{O}(h_x^3) = u_{2,j} + \mathcal{O}(h_x^2), \quad (21)$$

where the last equality uses $u_{2,j} = u_{1,j} + h_x u_x|_{1,j} + (h_x^2/2) u_{xx}|_{1,j} + \mathcal{O}(h_x^3)$ together with $u_x|_{1,j} = 0$. Hence the ghost node prescription $u_{0,j} = u_{2,j}$ is second-order consistent with the Neumann condition. Substituting into D_{xx} at $i = 1$ gives the modified boundary stencil

$$D_{xx} u_{1,j} = \frac{2(u_{2,j} - u_{1,j})}{h_x^2} = u_{xx}|_{1,j} + \mathcal{O}(h_x^2), \quad (22)$$

where the remainder retains second-order accuracy because both $u_{0,j}$ and $u_{2,j}$ are expanded about the same point x_1 using the vanishing first-derivative condition. The same argument applies at $i = M_x$ and at the y -boundaries.

These approximations are encoded in one-dimensional operator matrices. The discrete Laplacian in the x -direction is the tridiagonal matrix $\mathbf{L}_x \in \mathbb{R}^{M_x \times M_x}$ with entries

$$(\mathbf{L}_x)_{i,k} = \frac{1}{h_x^2} \begin{cases} -2 & \text{if } k = i, \\ 1 & \text{if } |k - i| = 1 \text{ and } 2 \leq i \leq M_x - 1, \\ 2 & \text{if } (i, k) = (1, 2) \text{ or } (i, k) = (M_x, M_x - 1), \\ 0 & \text{otherwise,} \end{cases} \quad (23)$$

where the off-diagonal value $2/h_x^2$ at the first and last rows arises from the ghost node substitution (21). The discrete gradient operator $\mathbf{G}_x \in \mathbb{R}^{M_x \times M_x}$ has the central difference stencil at interior nodes and zero rows at boundaries:

$$(\mathbf{G}_x)_{i,k} = \frac{1}{2h_x} \begin{cases} 1 & \text{if } k = i + 1 \text{ and } 2 \leq i \leq M_x - 1, \\ -1 & \text{if } k = i - 1 \text{ and } 2 \leq i \leq M_x - 1, \\ 0 & \text{if } i = 1 \text{ or } i = M_x, \\ 0 & \text{otherwise.} \end{cases} \quad (24)$$

The zero boundary rows ensure that the discrete gradient returns zero at the boundary, which is consistent with the Neumann condition to $\mathcal{O}(h_x^2)$ by virtue of (20) and $u_x|_{\partial\Omega} = 0$. The operators $\mathbf{L}_y \in \mathbb{R}^{M_y \times M_y}$ and $\mathbf{G}_y \in \mathbb{R}^{M_y \times M_y}$ are defined identically with h_y replacing h_x .

The passage to two dimensions exploits the tensor product structure of the Cartesian grid. Adopting column-major (lexicographic by x for fixed y) ordering, the grid function $u_{i,j}$ is stored as a vector $\mathbf{u} \in \mathbb{R}^M$ with $\mathbf{u}_{(j-1)M_x+i} = u_{i,j}$. By the properties of the Kronecker product, applying $\mathbf{L}_x$ independently along each horizontal grid line and summing with $\mathbf{L}_y$ applied along each vertical line produces the two-dimensional discrete Laplacian

$$\Delta_h = \mathbf{I}_{M_y} \otimes \mathbf{L}_x + \mathbf{L}_y \otimes \mathbf{I}_{M_x} \in \mathbb{R}^{M \times M}, \quad (25)$$

where $\mathbf{I}_{M_x}$ and $\mathbf{I}_{M_y}$ are identity matrices of the related sizes. The matrix Δ_h is sparse and symmetric with at most five nonzero entries per row (the standard five-point stencil). The two-dimensional gradient operators are constructed analogously:

$$\nabla_x^h = \mathbf{I}_{M_y} \otimes \mathbf{G}_x \in \mathbb{R}^{M \times M}, \quad \nabla_y^h = \mathbf{G}_y \otimes \mathbf{I}_{M_x} \in \mathbb{R}^{M \times M}. \quad (26)$$

The approximation properties of the one-dimensional operators carry over to the Kronecker construction, and the resulting spatial truncation error is controlled as follows.

Proposition 3.1 (Spatial truncation error). *Let $u \in C^4(\bar{\Omega})$ satisfy $\nabla u \cdot \mathbf{n}|_{\partial\Omega} = 0$. Then the discrete Laplacian (25) satisfies*

$$\Delta_h \mathbf{u} - \Delta \mathbf{u} = \mathbf{R}_h, \quad \|\mathbf{R}_h\|_\infty \leq \frac{h_x^2}{12} \|u_{xxxx}\|_{L^\infty(\Omega)} + \frac{h_y^2}{12} \|u_{yyyy}\|_{L^\infty(\Omega)}, \quad (27)$$

where $\Delta \mathbf{u}$ denotes the pointwise restriction of Δu to the grid. In particular, $\|\mathbf{R}_h\|_\infty = \mathcal{O}(h_x^2 + h_y^2)$.

Proof. At each interior node (x_i, y_j) with $2 \leq i \leq M_x - 1$ and $2 \leq j \leq M_y - 1$, the Kronecker structure of (25) yields $(\Delta_h \mathbf{u})_{i,j} = D_{xx} u_{i,j} + D_{yy} u_{i,j}$. By (18) and (19),

$$(\Delta_h \mathbf{u})_{i,j} - \Delta u(x_i, y_j) = \frac{h_x^2}{12} u_{xxxx}|_{\bar{x}_i, y_j} + \frac{h_y^2}{12} u_{yyyy}|_{x_i, \bar{y}_j},$$

which gives the stated bound. At boundary nodes, the modified stencil (22) retains $\mathcal{O}(h_x^2)$ accuracy as established in (21), so the bound extends to the full grid. $\square$

3.2. Semi-discrete system

Applying the spatial operators developed above to system (4) and using the reformulated EC equation (11), we obtain the semi-discrete system of $4M$ ordinary differential equations

$$\frac{d\mathbf{U}}{dt} = \mathbf{A}\mathbf{U} + \mathbf{N}(\mathbf{U}) + \mathbf{R}_h, \quad \mathbf{U}(t) = (\mathbf{C}(t), \mathbf{P}(t), \mathbf{I}(t), \mathbf{F}(t))^\top \in \mathbb{R}^{4M}, \quad (28)$$

where $\mathbf{C}, \mathbf{P}, \mathbf{I}, \mathbf{F} \in \mathbb{R}^M$ are the grid vectors for each variable, the spatial residual satisfies $\|\mathbf{R}_h\|_\infty = \mathcal{O}(h_x^2 + h_y^2)$ by Proposition 3.1 and the analogous estimates for the gradient operators (20), and the term $\mathbf{R}_h$ is incorporated into the semi-discrete formulation and dropped from the notation hereafter. The linear part is the $4M \times 4M$ block matrix

$$\mathbf{A} = \begin{pmatrix} d_C \Delta_h & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ k_4 \text{diag}(\mathbf{T}) & d_P \Delta_h - k_6 \mathbf{I}_M & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & d_I \Delta_h & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{pmatrix}, \quad (29)$$

where $\mathbf{T} \in \mathbb{R}^M$ is the vector of TAF values (6) evaluated at grid points. The nonlinear right-hand side is

$$\mathbf{N}(\mathbf{U}) = \begin{pmatrix} \mathbf{v}_x \odot (\nabla_x^h \mathbf{C}) + \mathbf{v}_y \odot (\nabla_y^h \mathbf{C}) + (\nabla \cdot \mathbf{v})_h \odot \mathbf{C} + k_1 \mathbf{C} \odot (\mathbf{1} - \mathbf{C}) \\ -k_3 \mathbf{P} \odot \mathbf{I} + k_4 \mathbf{T} \odot \mathbf{C} + k_5 \mathbf{T} - k_6 \mathbf{P} \\ -k_3 \mathbf{P} \odot \mathbf{I} \\ -k_2 \mathbf{P} \odot \mathbf{F} \end{pmatrix}, \quad (30)$$

where $\odot$ denotes the Hadamard product and $\mathbf{1} = (1, \dots, 1)^\top \in \mathbb{R}^M$. The discrete velocity field and its divergence, corresponding to the continuous expressions (12), take the form

$$\mathbf{v}_x = \alpha_2 \nabla_x^h \mathbf{I} - \alpha_1 \nabla_x^h \mathbf{F} - \alpha_3 \boldsymbol{\varphi}_x, \quad \mathbf{v}_y = \alpha_2 \nabla_y^h \mathbf{I} - \alpha_1 \nabla_y^h \mathbf{F} - \alpha_3 \boldsymbol{\varphi}_y, \quad (31)$$

$$(\nabla \cdot \mathbf{v})_h = \alpha_2 \Delta_h \mathbf{I} - \alpha_1 \Delta_h \mathbf{F} - \alpha_3 \Delta \boldsymbol{\varphi}, \quad (32)$$

where $\boldsymbol{\varphi}_x, \boldsymbol{\varphi}_y \in \mathbb{R}^M$ are the vectors obtained by evaluating the analytical derivatives of the auxiliary potential $\varphi(\mathbf{x}) = \alpha_4^{-1} \log(1 + \alpha_4 T(\mathbf{x}))$ at grid points, and $\Delta \boldsymbol{\varphi}$ is either precomputed analytically or approximated as $\Delta_h \boldsymbol{\varphi}$.

The formulation (28)–(30) makes explicit the splitting that motivates the time integration strategy: the linear operator $\mathbf{A}$ contains the stiff diffusion terms whose spectrum scales as $\mathcal{O}(h^{-2})$, while the nonlinear operator $\mathbf{N}$ contains the reaction kinetics and the advective flux whose Lipschitz constant scales as $\mathcal{O}(h^{-1})$ at most.

3.3. Strang splitting and time integration

In order to advance the semi-discrete system (28) from time $t_n = n\tau$ to $t_{n+1} = (n+1)\tau$, we employ Strang splitting [Strang \(1968\)](#), which symmetrically interleaves the reaction-advection and diffusion operators. Denoting by $\Phi_{\mathcal{R}}^s$ the flow map of the reaction-advection system $d\mathbf{U}/dt = \mathbf{N}(\mathbf{U})$ over a time interval of length s , and by Φ_D^s the flow map of the diffusion system

$$\frac{d\mathbf{U}}{dt} = \mathbf{A}_{\text{diff}}\mathbf{U}$$

with $\mathbf{A}_{\text{diff}} = \text{diag}(d_C \Delta_h, d_P \Delta_h, d_I \Delta_h, \mathbf{0})$, the Strang splitting advances the state as

$$\mathbf{U}^{n+1} = \Phi_{\mathcal{R}}^{\tau/2} \circ \Phi_D^{\tau} \circ \Phi_{\mathcal{R}}^{\tau/2} (\mathbf{U}^n). \quad (33)$$

This symmetric composition is second-order accurate in the splitting error [Hundsdorfer and Verwer \(2003\)](#): if both flow maps are computed exactly, $\mathbf{U}^{n+1} - \mathbf{U}(t_{n+1}) = \mathcal{O}(\tau^2)$ where $\mathbf{U}(t)$ is the solution of the unsplit system (28). In practice, each flow map is replaced by a numerical approximation.

Considering the reaction-advection substep, we employ the forward Euler method:

$$\Phi_{\mathcal{R}}^{\tau/2}(\mathbf{U}) \approx \mathbf{U} + \frac{\tau}{2} \mathbf{N}(\mathbf{U}), \quad (34)$$

which introduces a local truncation error of $\mathcal{O}(\tau^2)$ per half-step, and consequently a global error of $\mathcal{O}(\tau)$ over the full integration interval. Combined with the $\mathcal{O}(\tau^2)$ splitting error, the overall temporal accuracy is therefore $\mathcal{O}(\tau)$. Replacing the forward Euler integrator with the two-stage Heun method, which computes:

$$\tilde{\mathbf{U}} = \mathbf{U}^n + \frac{\tau}{2} \mathbf{N}(\mathbf{U}^n), \quad \mathbf{U}^* = \mathbf{U}^n + \frac{\tau}{4} [\mathbf{N}(\mathbf{U}^n) + \mathbf{N}(\tilde{\mathbf{U}})], \quad (35)$$

as predictor and corrector, respectively, we would recover $\mathcal{O}(\tau^2)$ local accuracy and thus $\mathcal{O}(\tau^2)$ global accuracy matching the splitting. The diffusion substep requires the solution of the implicit system

$$\Phi_D^{\tau}(\mathbf{u}^*) : \quad (\mathbf{I}_M - \tau d \Delta_h) \mathbf{u}^{**} = \mathbf{u}^* \quad (36)$$

for each diffusive variable $u \in \{C, P, I\}$ with coefficient d ; the ECM density F , having no diffusion, is unchanged during this substep ($F^{**} = F^*$). The direct solution of (36) would require factorizing an $M \times M$ sparse matrix with bandwidth $\mathcal{O}(M_x)$, costing $\mathcal{O}(M_x^2 M_y)$ operations.

3.4. ADI method of Peaceman–Rachford

We decompose the two-dimensional Laplacian (25) into its directional components,

$$\Delta_h = \mathcal{L}_x + \mathcal{L}_y, \quad \mathcal{L}_x = \mathbf{I}_{M_y} \otimes \mathbf{L}_x, \quad \mathcal{L}_y = \mathbf{L}_y \otimes \mathbf{I}_{M_x}, \quad (37)$$

where $\mathcal{L}_x$ acts only along horizontal grid lines and $\mathcal{L}_y$ only along vertical ones. The Peaceman–Rachford scheme [Peaceman and Rachford \(1955\)](#) replaces the monolithic implicit step (36) with two half-steps. The x -sweep is implicit in x and explicit in y :

$$(\mathbf{I}_M - \frac{\tau d}{2} \mathcal{L}_x) \mathbf{u}^{**} = (\mathbf{I}_M + \frac{\tau d}{2} \mathcal{L}_y) \mathbf{u}^*, \quad (38)$$

and the y -sweep reverses the roles:

$$(\mathbf{I}_M - \frac{\tau d}{2} \mathcal{L}_y) \mathbf{u}^{**} = (\mathbf{I}_M + \frac{\tau d}{2} \mathcal{L}_x) \mathbf{u}^*. \quad (39)$$

Eliminating $\mathbf{u}^{**}$ between (38) and (39) and comparing with the exact operator $\mathbf{I}_M - \tau d \Delta_h + (\tau d)^2 \mathcal{L}_x \mathcal{L}_y / 4$ shows that the ADI scheme approximates the Crank–Nicolson operator with an additional term $(\tau d)^2 \mathcal{L}_x \mathcal{L}_y / 4 = \mathcal{O}(\tau^2 h^{-4})$, which after multiplication by $\mathbf{u}$ contributes $\mathcal{O}(\tau^2)$ to the local error provided $\mathbf{u} \in C^4$. The scheme is therefore second-order in time for the diffusion substep and unconditionally stable, since the amplification factor is bounded by unity for all eigenmode combinations of $\mathbf{L}_x$ and $\mathbf{L}_y$.

The computational efficiency of the ADI method arises from the Kronecker structure of (37). Since $\mathcal{L}_x = \mathbf{I}_{M_y} \otimes \mathbf{L}_x$, the left-hand side of (38) factors as

$$\mathbf{I}_M - \frac{\tau d}{2} \mathcal{L}_x = \mathbf{I}_{M_y} \otimes \underbrace{\left(\mathbf{I}_{M_x} - \frac{\tau d}{2} \mathbf{L}_x \right)}_{=: \mathbf{T}_x \in \mathbb{R}^{M_x \times M_x}}, \quad (40)$$

which is block-diagonal with M_y identical tridiagonal blocks $\mathbf{T}_x$ of size M_x . The x -sweep therefore decouples into M_y independent tridiagonal linear systems, one for each horizontal grid line $j = 1, \dots, M_y$.

Denoting the mesh ratio $r_x = d\tau/(2h_x^2)$, the tridiagonal matrix $\mathbf{T}_x$ has the explicit coefficient structure

$$(\mathbf{T}_x)_{i,k} = \begin{cases} 1 + 2r_x & \text{if } k = i, \\ -r_x & \text{if } |k - i| = 1 \text{ and } 2 \leq i \leq M_x - 1, \\ -2r_x & \text{if } (i, k) = (1, 2) \text{ or } (i, k) = (M_x, M_x - 1), \\ 0 & \text{otherwise,} \end{cases} \quad (41)$$

where the doubled off-diagonal entries at the boundaries again reflect the Neumann ghost node construction. For each grid line j , the right-hand side of the x -sweep is obtained by applying the explicit operator $\mathbf{I}_{M_x} + (\tau d/2)\mathbf{L}_y$ to the column data. Writing out the y -stencil at node (i, j) with $r_y = d\tau/(2h_y^2)$, this gives

$$(\text{RHS})_{i,j} = \begin{cases} (1 - 2r_y)u_{i,j}^* + 2r_y u_{i,j+1}^* & \text{if } j = 1, \\ r_y u_{i,j-1}^* + (1 - 2r_y)u_{i,j}^* + r_y u_{i,j+1}^* & \text{if } 2 \leq j \leq M_y - 1, \\ 2r_y u_{i,j-1}^* + (1 - 2r_y)u_{i,j}^* & \text{if } j = M_y, \end{cases} \quad (42)$$

with the boundary cases arising from the Neumann condition $u_{i,0}^* = u_{i,2}^*$ and $u_{i,M_y+1}^* = u_{i,M_y-1}^*$. Each of the M_y resulting tridiagonal systems is solved in $\mathcal{O}(M_x)$ operations by the Thomas algorithm. The y -sweep proceeds identically with the roles of x and y exchanged. The total cost of one ADI step for a single variable is therefore $\mathcal{O}(M_y M_x + M_x M_y) = \mathcal{O}(M)$, which is optimal.

We note two properties of the tridiagonal matrix $\mathbf{T}_x$ that are relevant for the implementation. First, $\mathbf{T}_x$ is strictly diagonally dominant, since $|1 + 2r_x| > 2r_x \geq |-r_x| + |-r_x|$ at interior rows and $|1 + 2r_x| > |-2r_x|$ at boundary rows, ensuring that the Thomas algorithm is stable without pivoting. Second, $\mathbf{T}_x$ is the same for all M_y systems in a given sweep, so the forward elimination coefficients can be precomputed once and reused across the batch.

3.5. Explicit treatment of reaction and advection

The reaction-advection substeps (34) require the evaluation of $\mathbf{N}(\mathbf{U})$ defined in (30). The reaction terms are evaluated exactly at grid points and introduce no spatial discretization error.

The advective term $\mathbf{v} \cdot \nabla C$ in the first component of (30) requires the discretization of a first-order spatial derivative in the presence of a velocity field that may have large magnitude relative to the diffusion. We quantify this ratio by introducing the local mesh Péclet number:

$$\text{Pe}_h = \|\mathbf{v}\|_\infty h / d_C.$$

For the parameter regime of the angiogenesis model, Pe_h can exceed 10^2 on coarse grids. When $\text{Pe}_h > 2$, central differencing produces spurious oscillations, and upwind differencing is required.

The first-order upwind approximation to $v_x u_x$ at node (i, j) is

$$v_x^{i,j} u_x(x_i, y_j) = \begin{cases} v_x^{i,j} \frac{u_{i,j} - u_{i-1,j}}{h_x} + \mathcal{O}(h_x) & \text{if } v_x^{i,j} > 0, \\ v_x^{i,j} \frac{u_{i+1,j} - u_{i,j}}{h_x} + \mathcal{O}(h_x) & \text{if } v_x^{i,j} \leq 0, \end{cases} \quad (43)$$

and analogously for $v_y u_y$ in the y -direction. In order to quantify the truncation error precisely, consider the case $v_x^{i,j} > 0$. By Taylor expansion,

$$\frac{u_{i,j} - u_{i-1,j}}{h_x} = u_x|_{i,j} - \frac{h_x}{2} u_{xx}|_{i,j} + \mathcal{O}(h_x^2), \quad (44)$$

so the upwind approximation (43) satisfies

$$v_x^{i,j} \frac{u_{i,j} - u_{i-1,j}}{h_x} = v_x^{i,j} u_x|_{i,j} - \frac{h_x}{2} v_x^{i,j} u_{xx}|_{i,j} + \mathcal{O}(h_x^2).$$

The leading error term $-(h_x/2) v_x u_{xx}$ has the form of a numerical viscosity with coefficient $|v_x| h_x/2$. This is acceptable provided $|v_x| h_x/2 \ll d_C$, i.e., $\text{Pe}_h \ll 4$. Considering larger Péclet numbers, the numerical viscosity becomes significant but is preferable to the oscillatory behavior of central differencing.

The velocity components $\mathbf{v}_x, \mathbf{v}_y$ themselves are evaluated using the central gradient operators (26) applied to the current values of $\mathbf{I}$ and $\mathbf{F}$, together with the precomputed vectors $\boldsymbol{\varphi}_x$ and $\boldsymbol{\varphi}_y$, as given in (31)–(32). These evaluations introduce $\mathcal{O}(h^2)$ errors in $\mathbf{v}$, which propagate as $\mathcal{O}(h^2)$ perturbations in the advective flux.

The stability of the explicit substep is governed by the CFL condition

$$\tau \leq \frac{h_{\min}}{\|\mathbf{v}\|_{\infty}}, \quad h_{\min} = \min(h_x, h_y), \quad (45)$$

where $\|\mathbf{v}\|_{\infty} = \max_{i,j} \sqrt{(v_x^{i,j})^2 + (v_y^{i,j})^2}$. This is the only time-step restriction for the full scheme, since the diffusion is handled unconditionally by the ADI method. The ratio $\tau_{\text{ADI}}/\tau_{\text{explicit}} = \mathcal{O}(h^{-1})$ quantifies the advantage of the splitting approach, which becomes increasingly significant as the mesh is refined.

3.6. Discrete conservation and stability diagnostics

The discrete mass of a grid function $\mathbf{u} \in \mathbb{R}^M$ is defined via the composite trapezoidal quadrature rule on Ω_h :

$$\mathcal{M}_h(\mathbf{u}) = h_x h_y \sum_{i=1}^{M_x} \sum_{j=1}^{M_y} w_{i,j} u_{i,j}, \quad (46)$$

where the weights are $w_{i,j} = 1$ at interior nodes, $w_{i,j} = 1/2$ at boundary edges, and $w_{i,j} = 1/4$ at corners. This quadrature approximates the integral $\int_{\Omega} u \, d\mathbf{x}$ to $\mathcal{O}(h^2)$ for $u \in C^2(\bar{\Omega})$.

A key structural property of the discrete Laplacians $\mathbf{L}_x$ and $\mathbf{L}_y$ with Neumann boundary conditions is that their row sums vanish: $\mathbf{L}_x \mathbf{1} = \mathbf{0}$ and $\mathbf{L}_y \mathbf{1} = \mathbf{0}$. This follows directly from the matrix structure (23): summing row i gives $(-2 + 1 + 1)/h_x^2 = 0$ at interior rows, and $(-2 + 2)/h_x^2 = 0$ at boundary rows. Note that $\mathbf{L}_x$ is not symmetric, since the ghost-node construction yields $(\mathbf{L}_x)_{1,2} = 2/h_x^2 \neq (\mathbf{L}_x)_{2,1} = 1/h_x^2$, so the column sums do not vanish in general. However, defining the trapezoidal weight vector $\mathbf{w} = (1/2, 1, \dots, 1, 1/2)^T \in \mathbb{R}^{M_x}$, one verifies $\mathbf{w}^T \mathbf{L}_x = \mathbf{0}^T$. Since the quadrature (??) uses precisely these weights, it follows that $\mathbf{w}^T (\mathbf{I}_{M_x} - r_x \mathbf{L}_x)^{-1} = \mathbf{w}^T$, and the ADI diffusion substep preserves the discrete mass $\mathcal{M}_h(\mathbf{u})$ exactly. Through the quadrature weights (46), this means the diffusion substep does not alter the discrete mass, and any change in $\mathcal{M}_h$ is attributable solely to the reaction substep.

Taking into account the ECM density, which has no diffusion, this analysis is particularly clean. Applying the forward Euler reaction step to the fourth component of (30) gives $\mathbf{F}^{n+1} = \mathbf{F}^n - \tau k_2 \mathbf{P}^n \odot \mathbf{F}^n + \mathcal{O}(\tau^2)$, whence

$$\frac{\mathcal{M}_h(\mathbf{F}^{n+1}) - \mathcal{M}_h(\mathbf{F}^n)}{\tau} = -k_2 \mathcal{M}_h(\mathbf{P}^n \odot \mathbf{F}^n) + \mathcal{O}(\tau), \quad (47)$$

which is the discrete analog of the fourth law in (14).

The discrete energy functional

$$\mathcal{E}_h^n = \frac{h_x h_y}{2} \sum_{i,j} \left[d_C \left| (\nabla_h C)_{i,j}^n \right|^2 + d_P \left| (\nabla_h P)_{i,j}^n \right|^2 + d_I \left| (\nabla_h I)_{i,j}^n \right|^2 + (F_{i,j}^n)^2 \right] \quad (48)$$

provides a stability monitor for the scheme. In the absence of source terms, the diffusion substep is dissipative ($\mathcal{E}_h^{n+1} \leq \mathcal{E}_h^n$), while the reaction substep may increase the energy through the proliferation and production terms. In the context of a well-resolved simulation, the balance should yield $\mathcal{E}_h^{n+1} \leq \mathcal{E}_h^n + \mathcal{O}(\tau)$ when the source contributions are bounded.

CRedit authorship contribution statement

Pasquale De Luca: Conceptualization, Methodology, Software, Writing – Original Draft.

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